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The cereal network: a baseline approach to current configurations of trade communities

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Abstract

This paper attempts to provide insights into the current network configurations of the food-trade system and to study the short-term effects of one of the ongoing and lasting global crises, the Ukraine War, on the link intensity. Towards this end, this analysis (1) reveals the pattern of countries' network positions in two most traded sub-categories of the cereal network: *wheat and meslin*, and *maize or corn*, and (2) discusses the characteristics of the global cereal networks over the 2021–2022 period. The results highlight several features of the trade networks: (1) the distribution of cereal trade is highly concentrated, with considerable dependency on a small number of exporters and a low import diversification, making the system rigid and prone to shocks; (2) a central role of several key developed countries that leave many developing countries outside the centre of the networks; (3) a high network heterogeneity which confirms the propensity to have hub nodes. Particular indicators show that the highest level of interconnectivity is specific to the cereals' export network, the densest networks are the maize or corn ones, and the greatest heterogeneity appears for the cereals export network.

Keywords: Food-trade network, Food security, Export intensity index, Import intensity index, Cereal trade core countries, Network analysis, War in Ukraine

Introduction

In recent years, there has been a surge in research on the global food system as a network. The concepts of the food system are increasingly recognized as the basis for the development of policies to ensure food security and adopt more sustainable systems (Ercsey-Ravasz et al. 2012; Sartori, 2015; Brunori et al. 2016; Bene et al. 2019). However, the continuous dynamic of the characteristics of the food-system network is still poorly understood (Puma 2019). The rise in global food insecurity resulting from the COVID-19 pandemic due to restrictions on cereal exports (Falkendal et al. 2021), exacerbated by the war between Russia and Ukraine, in addition to existing risks (e.g. population growth (Jiang et al. 2014; van Dijk et al. 2021), rising food prices (Headey et al. 2016), agricultural production shocks (Ferguson and Gars, 2020), climate change (Fanzo et al. 2018)) determines an unprecedented pressure on the food system.

This context translates into future challenges to address the simultaneous increase in food demand and the need to move towards more sustainable food systems. The reliance

of diets on a small number of crops poses a threat to global food security if the production or trade of one or more of these crops is curtailed for any reason. Wheat, maize, and rice comprise approximately half of all human diets and represent 86% of all cereal exports (FAO 2021). The global food supply chain works as a highly integrated "just-in-time" delivery system, a method that saves money on storage costs, but any disturbance to this fragile system generates a snowball effect. Most countries depend on trade to meet domestic demand for staples (Puma et al. 2015), making the global food system vulnerable to both self-propagating risks (i.e. trade restrictions) and global systemic risks from shocks such as climate change or major crop diseases (Cottrell et al. 2019).

Applied to the study of food systems, the network analysis shows that crop trade networks, the volume of trade and the number of links between countries have increased during the last decades (Zhang and Zhou, 2022; Duan et al. 2022; FAO 2022). Interdependence between countries leads to a high degree of complexity in international trade (Burkholz, 2019), increases the possibility for countries to be influenced by the global market (Seekell et al. 2017), but also promotes the diversification of global supply risks (Burkholz et al. 2015). Studies such as Lin (2014), Torreggiani (2018), Hedlund (2022) have shown that agricultural trade is fragmented into trade communities or blocs, and countries (i.e. nodes) are expanding more and more rapidly their trade relations (i.e. links) within rather than outside their community.

We aim to contribute to this research by applying a baseline approach to provide insights into the current network configurations of the food-trade system and to study the effects of one of the ongoing and lasting global crises, the Ukraine War, on the link intensities. Towards this end, our analysis (1) reveals the pattern of countries' network positions in two most traded subcategories of the cereal network, *wheat and meslin*, and *maize or corn*, and (2) discusses the characteristics of the global cereal networks over the 2021–2022 period. The remaining sections provide a background to the global context of the cereal markets, introduce the methodological approach, discuss the results, and conclude with comments on the future challenges facing the cereal trade network.

Recent developments and the war in Ukraine

Negative influences on the agri-food markets were already in place at the time when the COVID-19 pandemic started in Spring 2020 (Andrés González-Moralejo 2024). "The prevalence of moderate or severe food insecurity" had become a worrying but familiar feature of global markets since 2014 (FAO et al. 2021, xii). However, the rate at which the commercial environment began to deteriorate was unprecedented. While some countries have imposed temporary export restrictions, others have reduced their barriers to trade, which determined a sharp increase in demand, mainly in the case of wheat (Gutiérrez-Moya et al. 2020). On the supply side, restrictive measures caused supply chain disruptions, followed by a rapid increase in prices. Between April 2020 and December 2021, wheat prices increased by 80 per cent (Thomé et al. 2022, xlv). Then, the military conflict began in February 2022 with Russia's invasion of Ukraine, involving two of the most important grain exporters and only amplifying massive food supply disruptions and price volatility. The impact of the war is so severe because Ukraine and Russia are top agricultural exporters—for example, Ukraine is the world's largest exporter of sunflower oil (50% of world exports), the third largest of barley (18%), the fourth largest of

maize (16%) and the fifth largest of wheat (12%) (European Council 2023)—and many low- or low-middle-income countries import more than 50 per cent of their wheat from Russia and Ukraine to feed their population (Oxfam 2022). Russia's and Ukraine's interruption of grain exports caused more than 50% reduction in direct imports to 30 countries, including Eritrea, Seychelles, Kazakhstan, and Mongolia (Liu et al. 2023). Household consumption has been imperiled in developing Asian, Middle Eastern, and African countries that rely heavily on imports of Ukrainian wheat — for instance Pakistan (for which, as of 2020, 49% of wheat imports originated from Ukraine), Lebanon (62% of wheat imports), and Egypt (23% of wheat imports) as wheat prices increased by 28% during the early phase of the war (Devadoss and Ridley, 2024).

Recent studies have been examining the potential economic repercussions of the conflict between Russia and Ukraine from various perspectives. Part of this literature has primarily focused on macroeconomic effects of the military conflict (Estrada and Koutronas, 2022; Mahlstein et al. 2022) or on financial impact on stock markets (Boubaker et al. 2022; Bounbou and Yatié, 2022; Umar et al. 2022; Evenett and Muendler, 2022; Mahlstein et al. 2022). Current studies also address the issue of shocks to food supply or food markets (IFPRI, 2022; Estrada and Koutronas, 2022; Just and Echaust 2022, Xu. et al., 2024). <https://doi.org/10.1186/s40100-024-00296-9>. Specific crop analysis addresses either the aggregated category of cereals (Dupas et al., 2019; Duan et al. 2022; Guan et al., 2022; Hellegers, 2022; Yin et al. 2023), one particular crop only such as wheat (Fair et al. 2017; Raj et al. 2022; Gutiérrez-Moya, 2020), corn and maize (Jayasinghe et al. 2010; Wu and Guclu 2013; Szerb et al. 2022) or consider wheat and maize together with other crops (Gang et al. 2023; Falkendal et al. 2021).

The conceptual framework of the paper is aimed at enriching the debate with an analysis of networks related to two major agricultural products, wheat and corn, whose markets were mostly disrupted by the Ukrainian conflict. The intensity index we use in our paper accounts for the number of trade links between countries, their trade value, and emphasizes the dynamic perspective of interdependence, albeit limited to the years before and immediately after the beginning of the war (2021–2022). In addition, we measure the centrality of nodes and networks, not only in terms of quantities and volumes, but also in relation to the main actors involved in the establishment of trade routes and connections.

Material and methods

Methods

A network analysis is a descriptive investigation that provides in-depth evidence regarding trade between countries, which are represented by nodes, and their bilateral volume of trade, as links with an associated length (i.e. strength of bilateral flows). As some links are shorter (more intensive trade), and some longer (less intensive trade), countries with similar length of the links belong to the same trade community.

We use the common set of indicators for the intensity of trade flows (e.g. Asia Regional Integration Center 2022) and estimate two indices—export intensity index (XII) and import intensity (III)—as follows:

Export intensity index—the ratio between the share of partner j in total reporter i 's exports and the share of world exports to partner j in the total world exports:

$$XII_{ij} = \frac{X_{ij}/X_{iw}}{X_{wj}/X_{ww}} \quad (1)$$

where X_{ij} is the export value of country i to country j , X_{iw} is the export value of country i to the world, X_{wj} represents the value of world exports to country j (imports of country j) and X_{ww} is the value of world exports.

Import intensity index, the ratio between partner j 's share in total reporter i imports and the share of world imports with partner j in total world imports.

$$III_{ij} = \frac{I_{ij}/I_{iw}}{I_{wj}/I_{ww}} \quad (2)$$

where I_{ij} is the import value of country i from country j , I_{iw} is the import value of country i from the world, I_{wj} represents the value of the world imports from country j (exports of country j) and I_{ww} is the value of world imports.

Finally, the two indicators have been normalized from 0 to 1 to maintain comparability between country pairs. If the normalized value is equal to 1, it indicates the strongest connection between country pairs. The formulas for normalization are as follows:

$$NXII = \frac{XII_{ij} - \min XII_{ij}}{\max XII_{ij} - \min XII_{ij}}; NIII = \frac{III_{ij} - \min III_{ij}}{\max III_{ij} - \min III_{ij}} \quad (3)$$

where $NXII$ represents the normalized export intensity index and $NIII$ represents the normalized import intensity index.

The compilation of both export and import networks allows the analysis of dependencies on certain partners, in specific numbers and to a certain extent for both exporters and importers. The two perspectives compose a more comprehensive picture when presented together.

We generate maps of trade networks using open-source Cytoscape software, which offers graphical layouts for nodes and edges (pairs of nodes) determined by certain algorithms. The *length* of the edge (represented as an arrow) is given by the normalized index values: the shorter the arrow, the higher the normalized intensity index, thus indicating a high intensity between the reporter and the partner, i.e. high mutual dependence. The colour intensity of the edge is directly proportional to the trade value of country i to country j (X_{ij}) and the trade value of country i from country j (I_{ij}), respectively. Arrows start from the partner and point to the reporting country. The two indices were calculated for each crop category.

Parameter selection

FAO (2021) uses the import dependency ratio ($IDR = \text{imports} \times 100 / (\text{production} + \text{imports} - \text{exports})$) to describe the dependence of a country on imports for a certain agricultural product. The intensity indices used in our paper have the advantage of considering the world share of each trade partner and emphasizing mutual dependence (the higher the intensity, the higher the dependence). For example, if the reporter is a major exporter to the partner, the XII_{ij} increases, but if the partner is a small importer relative to world, then XII_{ij} is higher compared to the case when the partner is a large importer relative to the world. Therefore, the vulnerability resulting from the

dependence of a reporter on the partner's market is reduced when the partner is a large importer relative to the world.

The nodes of the network display information related to centrality: the node diameter is directly proportional to the node *closeness centrality* and the colour of the node is darker for higher *betweenness centrality*. According to FAO (2022 a), the position and relative importance of a nation within the global food and agricultural trade network are shown by centrality measurements at national level. We use the *closeness centrality index*, which measures the proximity of a node to other nodes within the network based on the average shortest path length, indicating the rate at which information travels from one reachable node in the network to other nodes and it takes values from 0 to 1 (0 being the value for completely isolated nodes). On the other hand, the *betweenness centrality index* is used by FAO (2022 a) to measure the number of times a country connects to other nations that are not directly connected to each other. A high-value index is an indication of the emergence of trade-hub nations or trading communities. In Cytoscape, only networks without multiple edges are used to calculate betweenness centrality. The index range is also from 0 to 1 and it indicates the extent to which a node controls how other nodes interact with each other within the network. In dynamics, a declining centralization index may be a sign of transition to a more balanced trade network with strong international connections and decentralized trading arrangements. A complementary set of indicators for network analysis are briefly described in a methodological note in Appendix 1.

Data collection

We collected data for cereals (product code:10), wheat and meslin (product code:1001) and maize or corn (product code:1005) from the International Trade Center (Trade Map database). We selected data at the level of 2021 and 2022 to gauge the impact on global cereal networks before and after the start of the war in Ukraine (24 February 2022), a historical milestone that is supposed to weigh heavily on the future resilience of the global food system. To focus the analysis on relatively larger trade flows, bilateral trade flows below 60% of world trade (export and import) values were excluded, which means that the shares, the number, and identity of reporting countries may vary between maps. However, we still maintained a relatively high number of countries (cereal: 80 in 2021 and 75 in 2022, wheat 69 2021 and 65 in 2022, maize 50 in 2021 and 49 in 2022) in the dataset. The data used to build the maps are listed in separate tables available on request.

The data collection process for each product category followed several steps. Initially, we prioritized the reporters based on their share in world export/import. We then narrowed down our selection to those that covered at least 60%. If the combined share did not reach 60%, we added another country until it exceeded this threshold. The same procedure was conducted further for the partner countries. The analysis is limited to the most major players for several reasons. Firstly, this allows for a more focused analysis on the significant amounts involved in international trade with cereals. By excluding minor trade participants, we can avoid any misleading influence they may have on the formation of hubs. Additionally, representing only the main nodes and edges in the graphs improves their readability.

Results and discussion

Cereal export network

The maps in Appendix 2—Figures 1 and 2 (Cereal export network in 2021 and 2022, respectively) reveal the central position of seven countries that account for 62.37% of global cereal exports (see Table 1 for a synthetic view of central players in cereal exports and imports). In 2022, Ukraine drops from the fourth to the ninth position, accounting for 5.10% of world exports of cereals (a 25.71% decrease in value). Ukraine's rank position in 2021 is taken by Brazil in 2022, which is characterized by a similar centrality being positioned at a comparable distance to other nodes in the network.

The connections of *the USA*, which is the largest global exporter of cereals, to China, Mexico and Japan, as importers that account for 60% of its exports in 2021, are displayed in the darkest arrows (indicating the export volume) in both years, albeit at low intensities (long edges). Another key node is the *Russian Federation*, which has the highest export intensity with Kazakhstan in 2021, although Kazakhstan is Russia's seventh largest cereal importer (2.68%) and disappears from the main partners in 2022. Russia's relationship to Turkey becomes closer in 2022 when the export intensity grows. Russia's hub status changed from more to less important than the USA in 2021 compared to 2022. *India* is an interesting node because all its major importers are on the Asian continent, and even if most of them are in geographical proximity, it covers the entire range of export intensity values from high mutual dependence towards Bangladesh, Nepal, Benin, Iraq, Senegal, to very low towards China, Iran, and Vietnam. India's centrality remained constant and the most significant among exporters over the two-year period.

Wheat and meslin exports are represented in a more concentrated network (Appendix 2—Figures 5 and 6) as compared to total cereals, which means that network members are more dependent on a smaller number of exporters and thus more vulnerable to eventual shortages. In both years, the highest trade intensity was recorded for exports from Canada to the USA, having a normalized index of 1. Canada also remains the most important hub serving various markets on the American and Asian continent. In 2021, the *Russian Federation* was the largest exporter of wheat and meslin (\$7,301 m), representing 12.9% of global exports mainly to seven partners that purchased approximately

Table 1 Key players in cereal trade networks Source: Authors' calculations

	2021	2022
Export network ^a	USA (19.80%), Argentina (8.79%), India (8.01%), Ukraine (7.68%), Australia (6.51%), Russian Federation (5.95%), Canada (5.64%)	USA (17.50%), Argentina (8%), India (7.80%), Brazil (7.70%), Australia (7.70%), France (6.60%), Russian Federation (5.80%)
Import network ^b	China (11.96%), Iran (4.57%), Mexico (4.57%), Japan (4.52%), South Korea (3.01%), Egypt (2.96%), Turkey (2.55%), Vietnam (2.50%), Italy (2.47%), Indonesia (2.44%), Netherlands (2.43%), Spain (2.37%), Philippines (2.07%), Saudi Arabia (2.06%), Bangladesh (2.05%), Algeria (2.05%), Germany (1.99%), Brazil (1.73%), Belgium (1.70%), Nigeria (1.65%)	China (9.80%), Japan (4.90%), Mexico (4.50%), Iran (4.30%), South Korea (3.30%), Spain (3.30%), Egypt (3.20%), Italy (3.10%), Saudi Arabia (3.10%), Turkey (2.70%), Indonesia (2.20%), Germany (2.20%), Netherlands (2.20%), Philippines (2.20%), Algeria (2.00%), Morocco (1.90%), USA (1.70%), Colombia (1.70%), Brazil (1.70%)

^a seven large reporters that account for 62.37% of world exports (2021) and 61.10% (2022)

^b 20 countries cover more than 60% of world imports

60% of exports: Turkey, Egypt, Azerbaijan, Nigeria, Kazakhstan, Sudan, and Saudi Arabia. In 2022, only four countries accounted for 60% of its exports (Azerbaijan, Kazakhstan, Nigeria and Sudan disappearing from the list) and its centrality decreased severely. Ukraine dropped from the list of the first 60% exporters of wheat and meslin in 2022, while France remains the major European hub and its betweenness centrality index even increases in 2022.

Regarding *maize or corn* (Appendix 2—Figures 9 and 10), the node with the highest export value is *the USA*, accounting for approx. 35% of world trade in 2021 exporting mainly to China (26.68%), Mexico (25.02%), and Japan (16.75%). However, the USA does not hold a central position in the network and the mutual dependencies on its importers are relatively small. The next most important exporters in 2021 are *Argentina*, with trading partners spread over three continents (South America, Asia, and Africa), and *Ukraine* with 60% of its exports concentrated in relations with few importers (China, Spain, the Netherlands, Egypt, and Iran). In 2022, Brazil surpasses Argentina, both remaining major hubs and at a short distance to other nodes (as indicated by the betweenness and closeness centrality indexes). *Romania* (\$1936 m in 2021), next to *France* (\$1937 m in 2021), is an important European exporter and an important hub, with a positive evolution in 2022, when it also acted as an exit gate for Ukrainian grains. Ukraine's position has not been deteriorated neither in terms of export value nor node centrality. The *United Arab Emirates* (UAE) form an interesting node in 2021 because it has only one partner, Iran, and this bilateral relationship is of high mutual dependence having a normalized trade intensity index of 1.00. In 2022, their relationship is translated to the trade link between Argentina and Vietnam.

Cereal import network

The maps in Appendix 2—Figures 3 and 4 (Cereal import network in 2021 and 2022, respectively) now show the importers' network, revealing fewer nodes in 2021 (38 vs. 42 for exports) but more nodes in 2022 (39 vs. 36 for exports). At the same time, a greater number of edges than for exports in both years depicts a denser network for imports, especially in 2022. The number of countries which cover over 60% of world imports is almost the same in 2021 and 2022, but it is larger compared to exports (see Table 1), suggesting a relatively greater market power on behalf of exporters. China is the main importer of cereals, although its share in world imports decreases in 2022. It relies on a few partners, among which Ukraine severely diminishes its position in 2022. Notable dependencies are seen in the case of *Iran* (low diversity of import sources, large import volumes, and high intensity with the UAE plus Russia as a key supplier in terms of import value), which persist in 2022. Although *Turkey* had a high import concentration and reliance on two 'sensitive' markets such as the Russian Federation and Ukraine in 2021, the supplies increased from these countries in 2022. Given the values of the betweenness index, Russia becomes a more important hub in the region in 2022. At the European level, *France* represents a key node that supplies five other European markets together with Germany, which seems to act as a transitory node (it imports from France, Poland, and the Czech Republic and exports to the Netherlands, Belgium, and Nigeria).

Importers of wheat and meslin (Appendix 2—Figures 7 and 8) are mostly concentrated in Asia, in 2021, accounting for 28% of world imports. Bangladesh and India, Italy and Austria, Brazil and Argentina, Italy and Hungary have strong intensities in 2021, suggesting high interdependencies, especially between neighbouring countries, but which have greatly diminished in 2022. The network in both years is characterized by several hub nodes given the high betweenness centrality for Egypt, France, Ukraine, Italy, Russia, Canada, and Australia. In general, countries have few alternatives and rely on a small number of suppliers.

This conclusion also applies to *imports of maize or corn* where the maps (Appendix 2—Figures 11 and 12) show scattered regions of reporters (importers), with links to a small number of core exporters—the USA, Argentina, Ukraine, and Brazil. In Europe, Ukraine remained a moderately important exporting hub and supplies significant amounts of maize or corn to Spain, the Netherlands and Italy, although the three economies, and the European countries in general, are not highly dependent on Ukrainian supplies.

Discussion

Some technical aspects of network analysis are worth mentioning. Analysing the indicators that characterize each network (Table 2), we find that in the case of *cereals*, the number of nodes and edges is the largest, which is easy to anticipate because it comprises several product categories, including *wheat and meslin* and *maize or corn*. However, in 2022, the number of nodes and edges significantly decreased especially in the case of total cereals exports and wheat and meslin exports and slightly increased in the import networks for the same goods. This suggests a higher concentration of exports and the disappearance of certain players above the breakpoint in 2022.

The highest average number of neighbours is for the import of total cereals and wheat and meslin, with a small decrease in 2022, showing that each node has more connections (they do not rely on a limited number of partners). A similar situation is revealed for the import of the same goods, where a high characteristic path length suggests many indirect connections through other trade partners. While in most cases there was a decrease from 2021 to 2022 for this index, a reverse trend can be observed for wheat and meslin export and cereals import.

Among the three product categories, the highest values of the network density index are for the *maize or corn* networks (export and import) over the period. Although the number of nodes and edges is the smallest for this product, a higher network density reduces the dependency. The import of *cereals* in 2021 had the highest degree of interconnectedness, as indicated by the cluster coefficient, but in 2022 it decreased significantly, when the first place was occupied by the export of *wheat and meslin*. The high heterogeneity of the network confirms the propensity to have hub nodes that may translate into trading communities. For the export of *cereals*, the heterogeneity index had its maximum value in 2021 with a slight decrease in 2022..

Conclusions

In line with recent studies (Caparas et al. 2021, Zhang and Zhou 2022; Duan et al. 2022), our results point to a highly concentrated cereal network. Consequently, the dependency on a small number of exporters and a low diversification of imports lead to a global

Table 2 Networks analysis indicators Source: Authors' calculations

Indicator	Export				Import							
	Cereals		Wheat and meslin		Maize or corn		Cereals		Wheat and meslin		Maize or corn	
	2021	2022	2021	2022	2021	2022	2021	2022	2021	2022	2021	2022
No. of nodes*	42	36	33	27	26	26	38	39	36	38	24	23
- Reporters	7	7	6	5	7	6	20	19	19	19	11	12
- Partners	35	30	28	23	19	21	20	23	16	21	13	11
No. of edges	54	49	42	31	33	33	56	57	47	48	28	27
Average no. of neighbours	2.57	2.72	2.54	2.30	2.53	2.54	2.94	2.92	2.70	2.52	2.47	2.48
Characteristic path length	3.21	2.96	3.15	3.57	3.19	3.11	3.43	3.56	3.64	3.59	3.12	2.92
Clustering coefficient	0.01	0.04	0.03	0.05	0.00	0.03	0.07	0.02	0.00	0.00	0.00	0.00
Network density	0.06	0.08	0.08	0.09	0.10	0.10	0.08	0.08	0.08	0.07	0.12	0.12
Network heterogeneity	1.04	0.99	0.93	0.95	0.76	0.81	0.66	0.65	0.57	0.67	0.53	0.72
Network centralization	0.21	0.31	0.24	0.28	0.19	0.24	0.14	0.14	0.10	0.13	0.13	0.31

Since certain countries may appear in both roles—as partners and reporters—and because each country is counted only once as a node, the total number of nodes may not equal the total number of reporters and partners

cereal-system vulnerable to shocks. In 2022, the networks became even more concentrated, as the number of nodes decreased, yet more connections were created for each node. Additionally, the centralization indices have generally increased in 2022, pointing to new or more dominant hubs.

Against this background, the global food system, for which cereal trade represents a crucial component, is not robust and may be easily affected by self-propagating disruptions, considering the interdependencies between countries.

As it happens, future scenarios are built not only on economic assumptions (e.g. by what extent the major grain producers such as China, India, the USA, Canada, France, and Germany could increase their export to alleviate the global wheat crisis (Lin et al. 2023), or on exogenous determinants (e.g. long-term (2070–2099) climate change impacts on food production (Hedlund et al., 2022), but increasingly also on geopolitical simulations. For example, the most important importers of Russian cereals are Turkey and Egypt, but Russia is also a key cereal provider for two members of the Commonwealth of Independent States (Azerbaijan and Kazakhstan), registering a high intensity index, which indicates a high mutual dependence. In this sense, there is a low probability for Russia to be affected by trade sanctions imposed by Western allies because of its invasion of Ukraine. Furthermore, as many variables impacting the cereal network will remain persistent and rather challenging to mitigate (e.g. climate change, trade sanctions, crises or conflicts of various nature), the diagnosis raises the question of solutions to make dependency relations more flexible, a concern much highlighted in the case of conflicts involving key actors.

A possible solution to manage these vulnerabilities, as Puma et al. (2015) suggest, would be to promote redundancy and diversity, which aim to compensate the potential losses of production/exports if trade is interrupted in a region. Redundancy improvements might focus on increases in domestic production or diet diversification to help mitigate dependence on main crops, wheat, and maize. On the supply side, due to location advantages, only a small number of countries specialize in the production of cereals for export, on which many other countries depend, including many of the poorest and most food-insecure countries. As one study remarked, it is a paradox of the global agricultural landscape that countries with high biophysical capacity do not have the ability to transform agricultural systems, given the weakness of their institutional framework, the lack of capital, infrastructure, and know-how and sometimes, even resistance to change (Seekell et al. 2017). This leads to an unmined production and trade potential of these countries that calls for intervention from both local and international actors. At national level, policy recommendations address the increasingly diverse functions of agriculture, from a sector that has been primarily associated with its role in providing food to its contributions to rural development and environmental protection. Complementarily, an international mechanism of coordinated financial, institutional and managerial support designed to develop their productive and export capacities is highly important.

On the demand side, the problem of concentration reveals that the global food system is rigid and inflexible in the face of numerous types of shock, making it prone to volatility. In this matter, the concept of countervailing power in trade can play a crucial role in addressing the dominance of large agricultural producers and preventing price manipulation. This could translate into an institutional mechanism to organize the market so

that import-dependent countries could offset the large marketing power of the dominant exporters. This can create a more even playing field for smaller cereal-producing nations, promote fair competition and facilitate market access for cereal-producing nations as well as for import-dependent countries to ensure a more sustainable and resilient agricultural sector.

The network analysis performed in this paper emphasizes current alternatives for both supplying and destination markets and the reconfiguration of trade networks within one year after receiving a major shock. A limit of our research is that the analysis does not account for re-exports and the likelihood that certain imports may originate from other countries, such as Russia or Ukraine, posing indirect food security problems. According to some estimates, ca. 20% of all agri-food exports were re-exported by the first importing country in 2014 (OECD 2020, 7), while in the case of Australia, 20% of its agricultural exports are re-exported by China as part of their own agri-food exports (OECD 2020, 20). Therefore, to arrive at more robust results, further research should consider the value of re-exports which apparently plays a non-negligible role. Another limitation derives from by the nature of network analysis itself which offers a snapshot of the links existent at certain moments in time and cannot be used as a tool to evaluate the effects of certain determinants or for making predictions, aspect which was beyond the scope of our research.

Appendix 1

We use a set of indicators to describe the network maps in Table 2. The *degree* is the number of neighbours for a node. The *average number of neighbours* or the average degree characterizes the entire network. Two nodes can be linked directly or by other nodes and edges following a certain path. The number of edges that make up a path determines the path length. The *characteristic path length* provides the expected distance between two connected nodes. The *clustering coefficient* measures the level of interconnectivity of neighbouring nodes.

In directed networks, the clustering coefficient is calculated using the following formula $C_n = e_n / (k_n(k_n - 1))$, where e_n is the number of pairs connected between all neighbours of node n and k_n is the number of neighbours of n . None of the node's neighbours are neighbours to one another if it equals 0, and all of the node's neighbours are neighbours to one another if it equals 1. The *network density* is a normalized form of the average number of neighbours, which reveals the typical connectedness of a network node and displays how numerous the network's edges are (a number between 0 and 1 represents the density self-loops and duplicated edges are ignored). The *network heterogeneity* shows the network's propensity to have hub nodes. Finally, we analyse the *centralization* at network level. Decentralized networks are defined by having a network centralization index close to 0, while star-shaped networks have a network centralization index close to 1.

Appendix 2

See Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12.

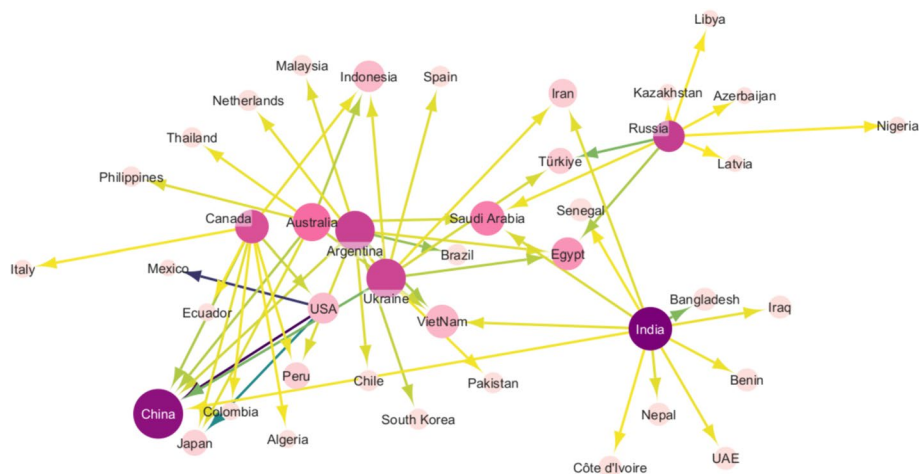


Fig. 1 Export intensity index for cereals 2021 Source: Authors' work using Cytoscape

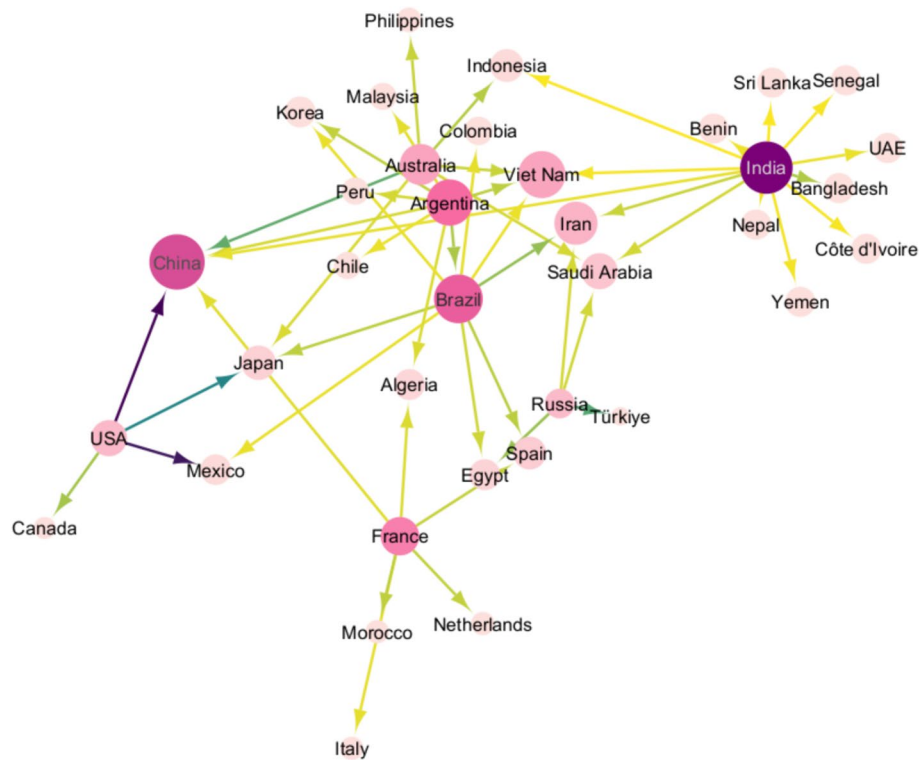


Fig. 2 Export intensity index for cereals 2022 Source: Authors' work using Cytoscape

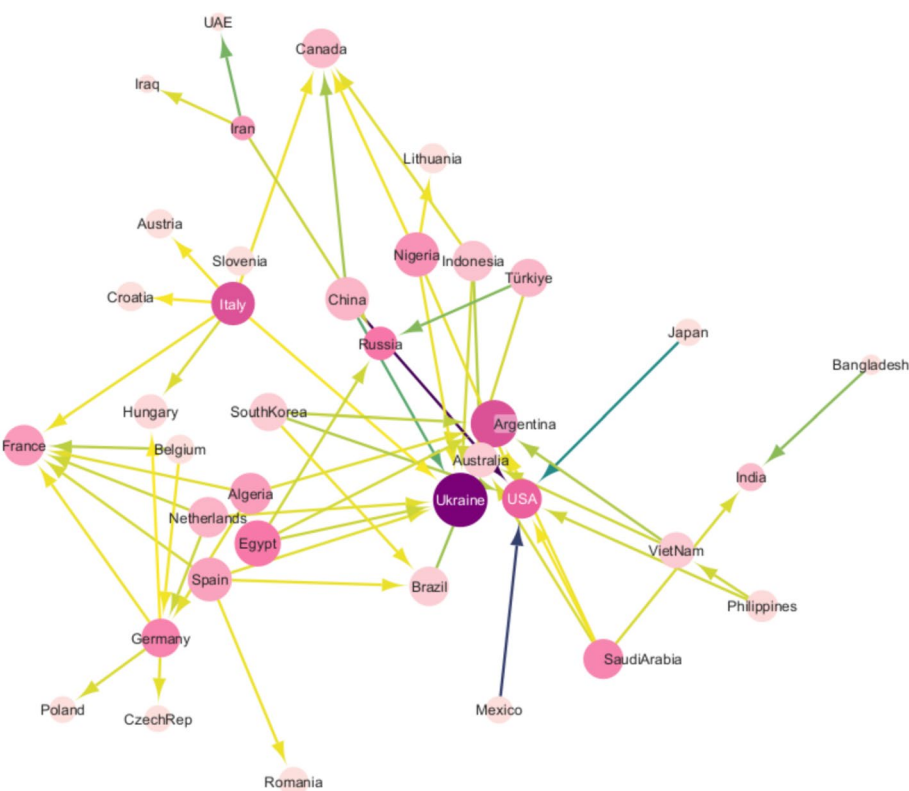


Fig. 3 Import intensity index for cereals in 2021 Source: Authors’ work using Cytoscape

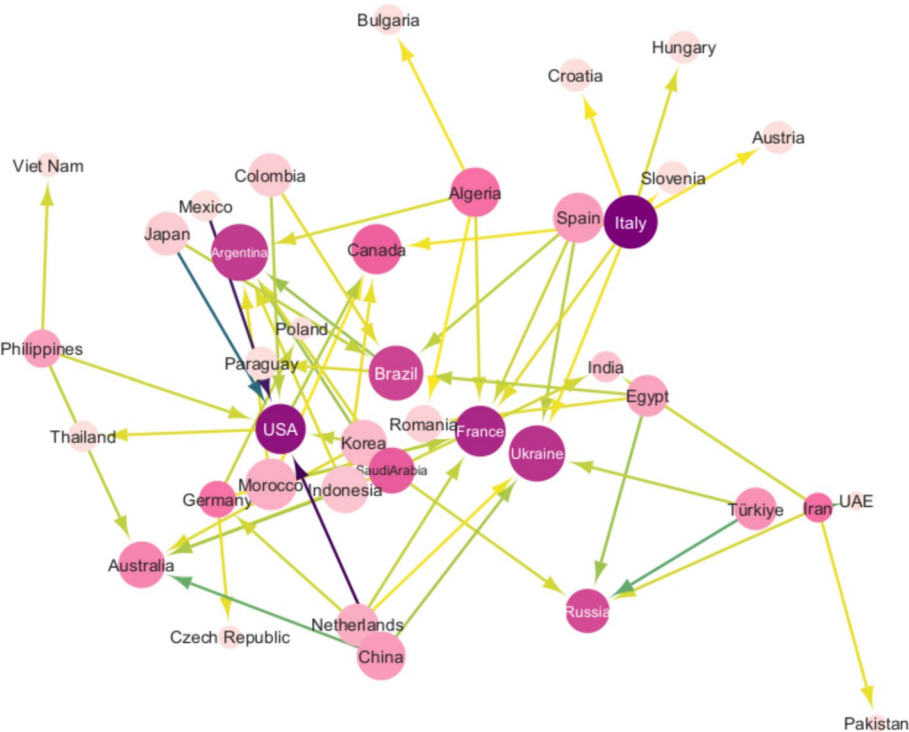


Fig. 4 Import intensity index for cereals in 2022 Source: Authors’ work using Cytoscape

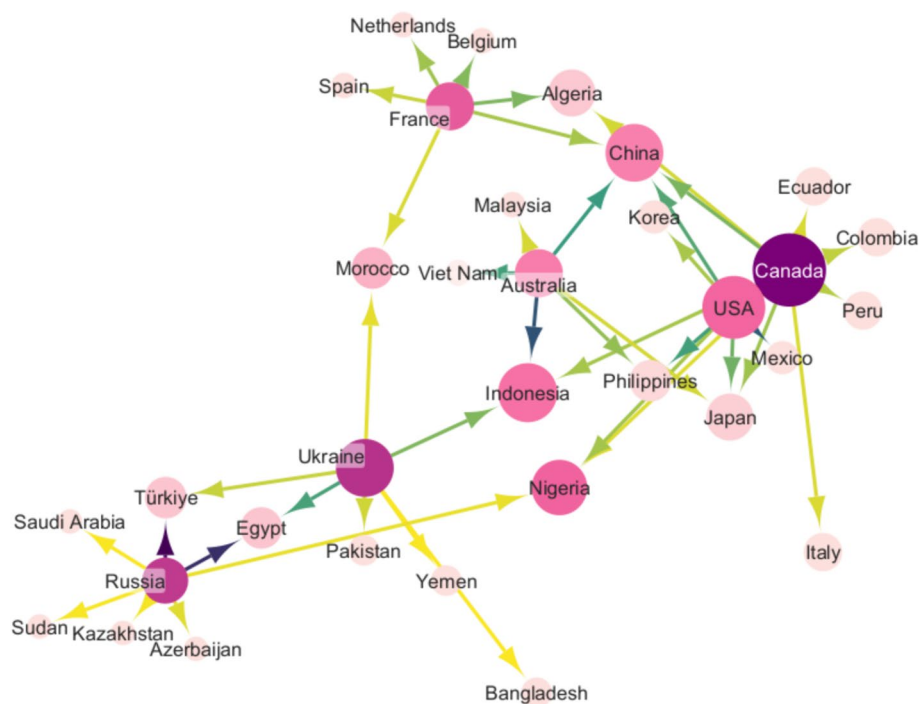


Fig. 5 Export intensity index for wheat and meslin in 2021 Source: Authors' work using Cytoscape

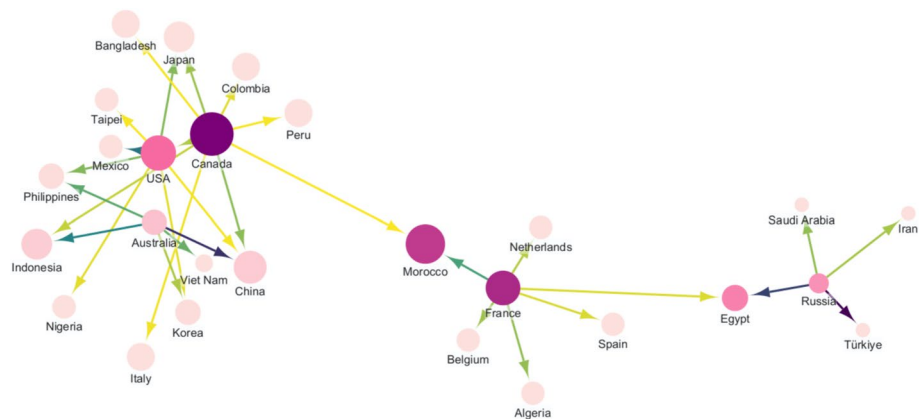


Fig. 6 Export intensity index for wheat and meslin in 2022 Source: Authors' work using Cytoscape

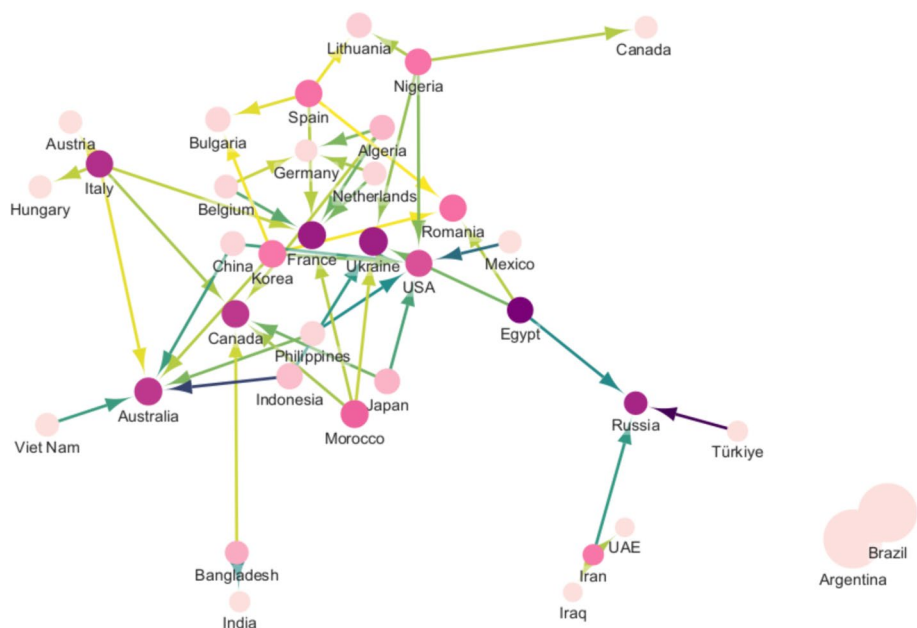


Fig. 7 Import intensity index for wheat and meslin in 2021 Source: Authors' work using Cytoscape

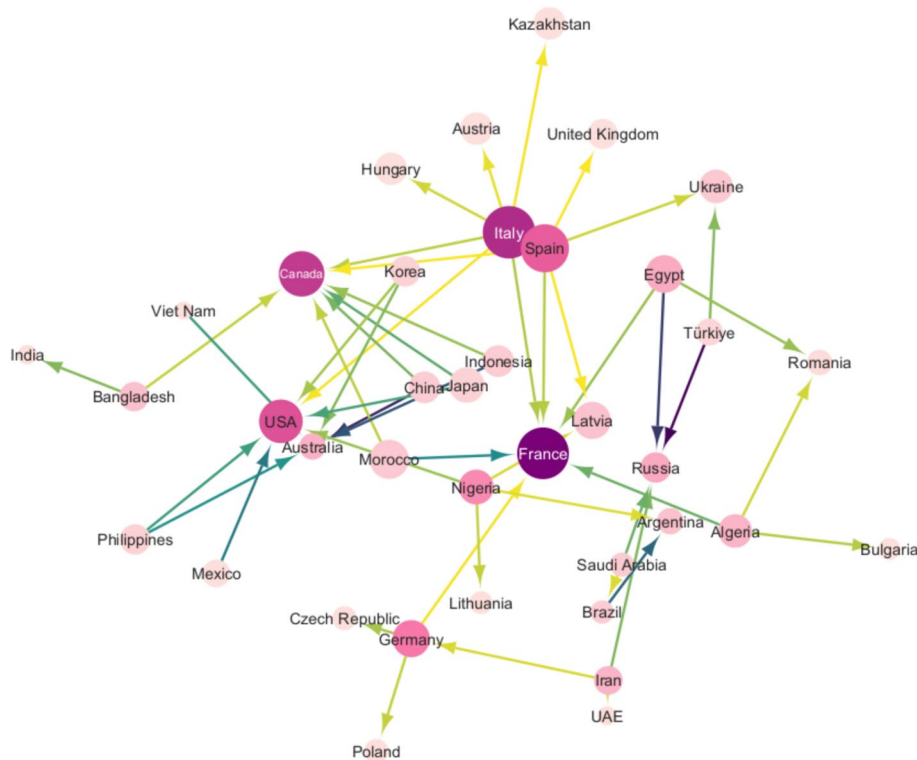


Fig. 8 Import intensity index for wheat and meslin in 2022 Source: Authors' work using Cytoscape

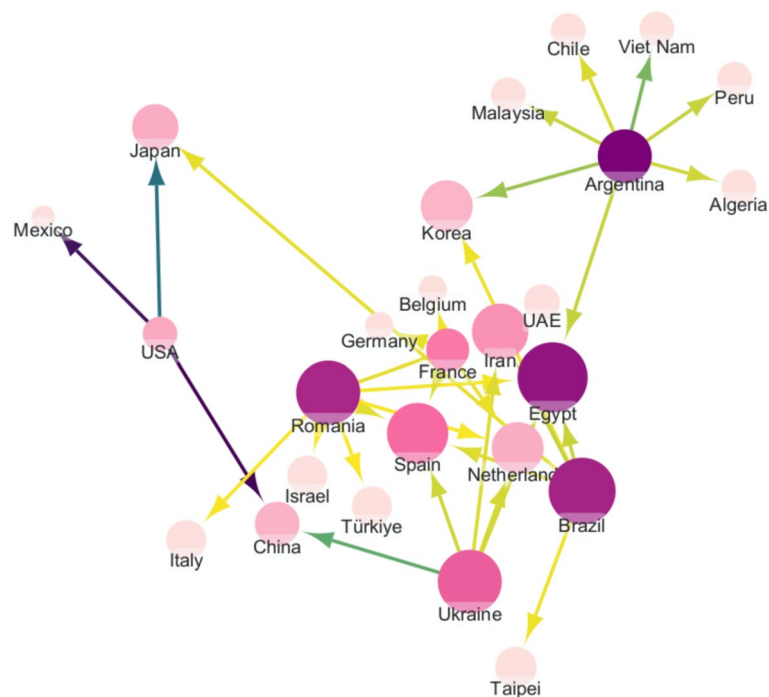


Fig. 9 Export intensity index for maize or corn in 2021 Source: Authors’work using Cytoscape

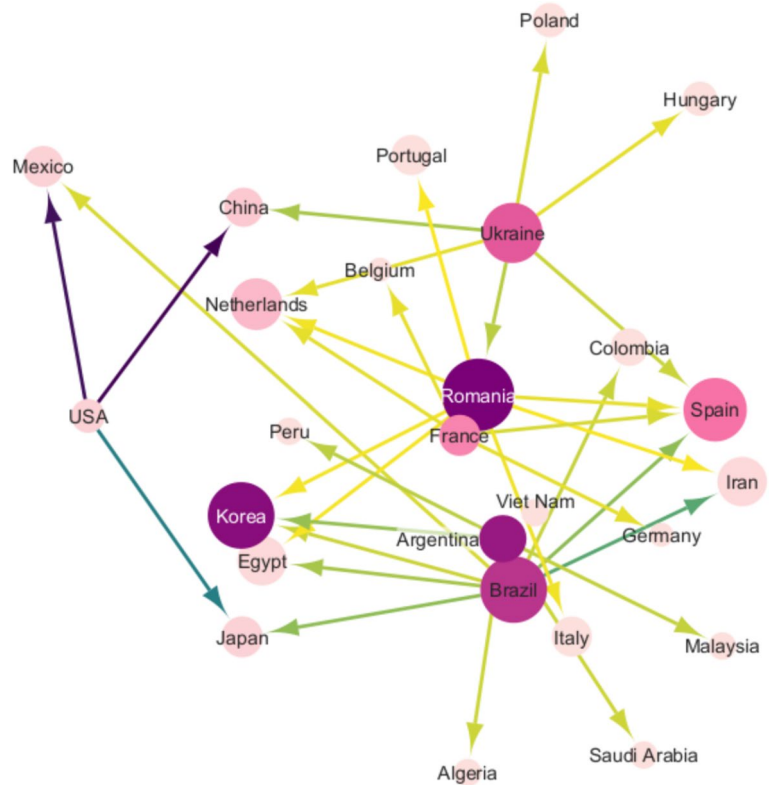


Fig. 10 Export intensity index for maize or corn in 2022 Source: Authors’work using Cytoscape

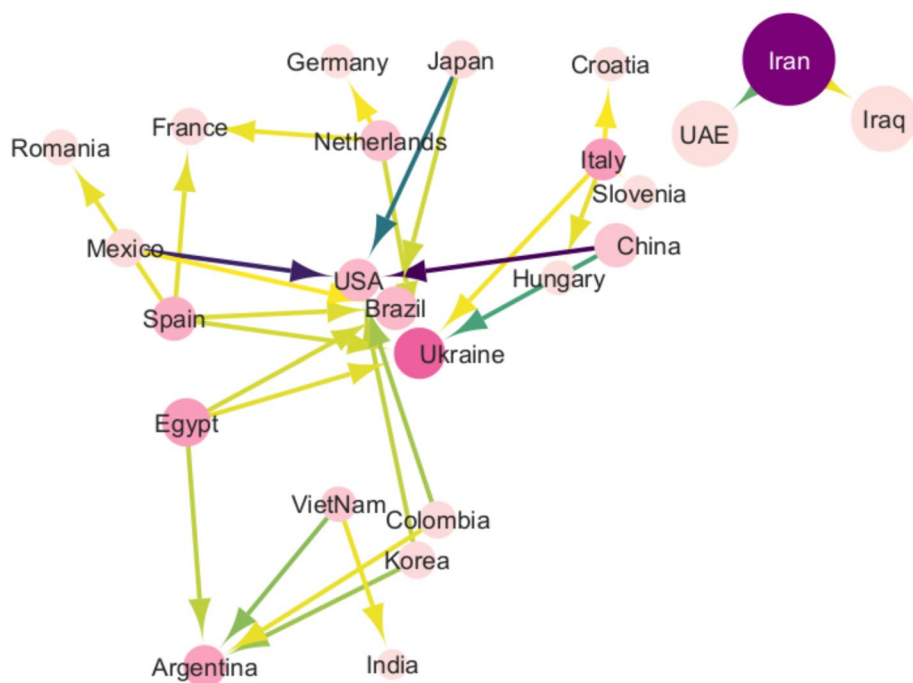


Fig. 11 Import intensity index for maize or corn in 2021 Source: Authors’work using Cytoscape

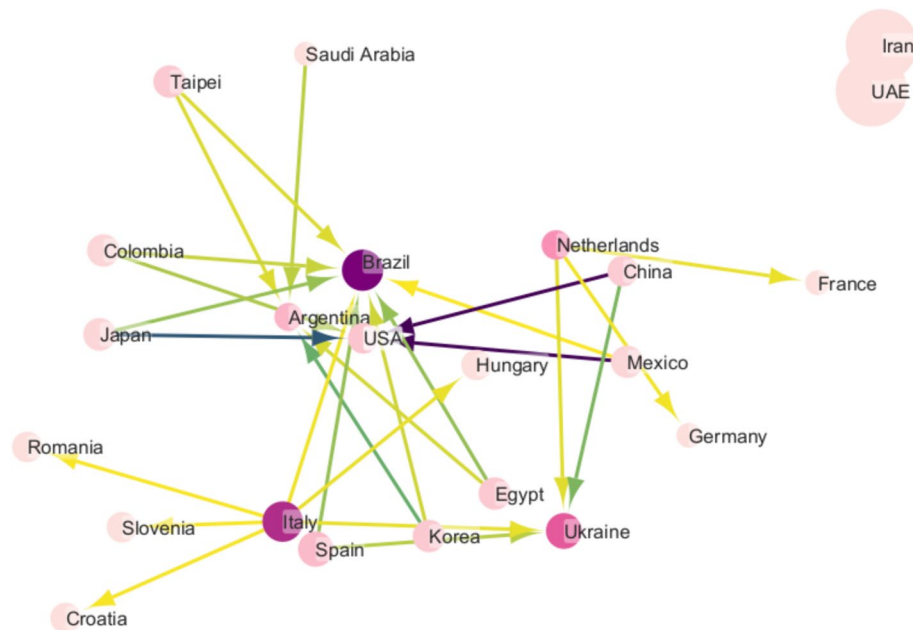


Fig. 12 Import intensity index for maize or corn in 2022 Source: Authors’work using Cytoscape

Abbreviations

EU	European Union
FAO	Food and Agriculture Organization
IDR	Import dependency ratio
IFPRI	The International Food Policy Research Institute
III	Import intensity index
Iij	The trade value of country i from country j

MENA countries	Middle East and North Africa countries
NAFTA	North American Free Trade Agreement
NIII	Normalized import intensity index
NXII	Normalized export intensity index
UAE	United Arab Emirates
USA	United States of America
XII	Export intensity index
Xij	The trade value of country i to country j

Acknowledgements

Not applicable

Author contributions

VC, APA and RGR have conceived the study and were responsible for the design and development of the data analysis. RGR and APA were responsible for data collection and analysis. VC has been responsible for data validation. RGR and APA were responsible for data interpretation. VC, APA, RGR, and DM wrote the first draft of the article. VC and DM were responsible for manuscript supervision. All authors read and approved the final manuscript.

Funding

Not applicable.

Availability of data and materials

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare that they have no competing interests.

Received: 21 April 2023 Revised: 20 February 2024 Accepted: 28 June 2024

Published online: 12 July 2024

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